

Exact Common Information and Exact Channel Synthesis for Correlated Gaussian Sources

Lei Yu

Abstract

In this paper, we resolve two conjectures posed by Yu and Tan in 2020 (in two separate papers published in the IEEE Trans. Inf. Theory). Specifically, we establish that: 1) the exact common information for a pair of ρ -correlated Gaussian sources is given by the conjectured expression $\frac{1}{2} \log \frac{1+\rho}{1-\rho} + \frac{\rho}{1+\rho}$; and 2) the admissible region for the shared randomness rate and the communication rate in exact channel synthesis is exactly the conjectured one. These results yield two important consequences. First, for any $\rho > 0$, the exact common information of a correlated Gaussian pair strictly exceeds Wyner's common information. Second, for $\rho > 0$, the exact channel synthesis of such a pair requires strictly higher rates than the total-variation version. The proof combines an exact optimal-transport representation of the worst-case Gaussian cross-entropy, Fathi's Gaussian transport inequality, and a determinant inequality arising from the covariance structure of the conditional means.

Index Terms

Gaussian sources, Exact common information, Exact channel synthesis, Communication complexity, Transport inequality

I. INTRODUCTION

A. Common information

The common information problem asks for the minimum amount of common randomness needed to generate two correlated random variables at two separate terminals. In Wyner's formulation [1], Wyner considered approximate generation in relative entropy and obtained the common information

$$C_{\text{Wyner}}(X; Y) = \inf_{W: X-W-Y} I(X, Y; W).$$

The exact common information problem, introduced by Kumar, Li, and El Gamal [2], is different. Here the synthesized distribution must equal the target distribution exactly, rather than merely approach it asymptotically in a weak metric. The common random variable is first

L. Yu is with the School of Statistics and Data Science, LPMC, KLMDASR, and LEBPS, Nankai University, Tianjin 300071, China (e-mail: leiyu@nankai.edu.cn). This work was supported by the National Key Research and Development Program of China under grant 2023YFA1009604 and the NSFC under grant 62101286.

generated and then independently processed at the two terminals. The exact common information is the minimum asymptotic rate of this common randomness.

Let π_{XY} be a probability distribution on a standard Borel product space. At blocklength n , an exact synthesis code consists of a discrete random variable W and conditional distributions $P_{X^n|W}, P_{Y^n|W}$, such that

$$P_{X^n Y^n} = \sum_w P_W(w) P_{X^n|W=w} P_{Y^n|W=w} = \pi_{XY}^{\otimes n},$$

where the conditional independence relation $X^n - W - Y^n$ holds.

The exact common information is the asymptotic normalized entropy of the smallest such common random variable. The variable-length formulation is equivalent because for a prefix-free code, $H(W) \leq L(W) < H(W) + 1$, and therefore the difference is negligible after normalization by n . Accordingly,

$$C_{\text{Exact}}(\pi_{XY}) = \lim_{n \rightarrow \infty} \frac{1}{n} \inf \{ H(W) : P_{X^n Y^n} = \pi_{XY}^{\otimes n}, X^n - W - Y^n \}. \quad (1)$$

A natural interpretation of exact common information can be described via the following thought experiment. Consider a random particle (or planet) that, at a certain instant, decomposes into multiple components. These components inherit shared common randomness and then evolve independently, governed jointly by this common randomness as well as their respective individual randomness. After some time has elapsed, given the observed joint distribution of the components, one may wish to estimate the amount of common randomness they possess. Or conversely, given the quantity of common randomness shared among them, we aim to characterize the set of feasible joint distributions these components can attain. This is exactly the common information problem.

The exact common information is no smaller than Wyner's common information. However, unlike Wyner's common information, the single-letter characterization of exact common information is rarely known, except for doubly symmetric binary sources (DSBSes) [3] and a certain class of sources satisfying $C_{\text{Exact}} = C_{\text{Wyner}}$ [2], [4], [3]. As for Gaussian sources, the exact common information is still unknown.

In addition, Yu and Tan [5], [6], [3] introduced the notion of Rényi common information, which is defined as the minimum common rate when the relative entropy is replaced by more general divergences—the family of Rényi divergences. The family of Rényi common information unifies Wyner's common information and exact common information since it includes them as two special cases with the Rényi order equal to 1 or ∞ .

B. Distributed channel synthesis

Common information has natural applications in distributed channel synthesis [7], [8], [9], [10], [11]. The latter problem, illustrated in Fig. 2, refers to the problem of determining the minimum communication rate required to generate a bivariate source $\{(X^n, Y^n)\}_{n \in \mathbb{N}}$ with X^n generated at the encoder and Y^n generated at the decoder such that the induced joint distribution $P_{X^n Y^n}$ approximately or exactly equals $\pi_{XY}^{\otimes n}$ for all $n \in \mathbb{N}$. In this paper, we focus on the exact

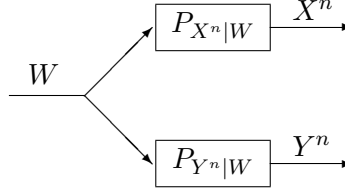


Fig. 1. Distributed source synthesis.

synthesis, equivalently, requiring $P_{X^n Y^n} = \pi_{XY}^{\otimes n}$. When there is no shared randomness, the exact channel synthesis problem reduces to the exact common information problem.

Consider the distributed source simulation setup depicted in Fig. 2. A sender and a receiver share a uniformly distributed source of randomness¹ $K \sim \text{Unif}(\mathcal{K})$, $\mathcal{K} := [e^{nR_0}]$. The sender has access to a memoryless source $X^n \sim \pi_X^{\otimes n}$ that is independent of K_n , and wants to transmit information about the correlation between correlated sources $(X^n, Y^n) \sim \pi_{XY}^{\otimes n}$ to the receiver. Given the shared randomness and the correlation information from the sender, the receiver generates a memoryless source $Y^n \sim \pi_{Y|X}^{\otimes n}(\cdot|X^n)$. Specifically, given X^n and K , the sender generates a “message” (i.e., a discrete random variable) M by a random mapping $P_{M|X^n K}$, and then sends it to the receiver error free. Upon accessing to K and receiving M , the receiver generates a source Y^n by a random mapping $P_{Y^n|MK}$. The joint distribution induced by this code is

$$P_{X^n K M Y^n} := P_{X^n} P_K P_{M|X^n K} P_{Y^n|MK}.$$

Now we would like to determine the minimum amount of communication such that $P_{X^n Y^n} = \pi_{XY}^{\otimes n}$ (or equivalently, $P_{Y^n|X^n} = \pi_{Y|X}^{\otimes n}$).

Definition 1. The admissible region of shared randomness rate and communication rate for the exact channel synthesis problem is defined as

$$\begin{aligned} & \mathcal{R}_{\text{Exact}}(\pi_{XY}) \\ &= \text{cl} \bigcup_{n \geq 1} \left\{ (R_0, R) : \exists (P_{M|X^n K}, P_{Y^n|MK}) \text{ s.t. } \right. \\ & \quad \left. \begin{aligned} & P_{Y^n|X^n} = \pi_{Y|X}^{\otimes n}, \\ & R \geq \frac{1}{n} H(M|K) \end{aligned} \right\}. \end{aligned} \quad (2)$$

In contrast to exact channel synthesis, total variation (TV)-approximate synthesis only requires the TV distance between the empirical distribution $P_{X^n Y^n}$ and the target distribution $\pi_{XY}^{\otimes n}$ to vanish asymptotically. Early studies by Bennett et al. [7] and Winter [8] investigated exact and TV-approximate channel synthesis under the assumption of unlimited shared encoder–decoder randomness, proving that the minimal asymptotic communication rate for both synthesis schemes equals the mutual information $I(X; Y)$ of the target distribution $(X, Y) \sim \pi_{XY}$.

¹For simplicity, we assume that e^{nR_0} is integers.

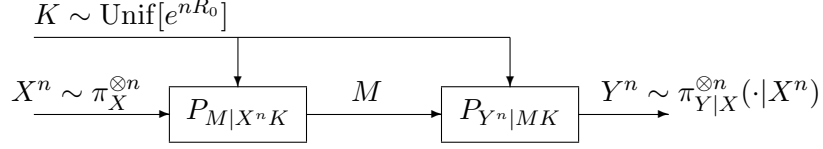


Fig. 2. Exact channel synthesis.

Subsequent research explored the fundamental tradeoff between communication rate and shared randomness rate in TV-approximate synthesis. Cuff [9] and Bennett et al. [10] characterized this rate tradeoff, while Harsha et al. [11] adopted a rejection sampling framework to analyze one-shot exact synthesis for discrete sources, establishing a bounded shared randomness cost with a mild increment in expected description length. Li and El Gamal [12] further refined the finite shared randomness upper bound for discrete settings with a negligible communication rate penalty.

Most prior work focused on TV-approximate synthesis or exact synthesis under extreme randomness conditions, except for [13]. Yu and Tan [13] fully characterized the optimal tradeoff between shared randomness rate and communication rate for exact synthesis of the doubly symmetric binary source (DSBS), and verified that exact synthesis requires a strictly higher communication rate than its TV-approximate counterpart. This is the first example for which the optimal rate tradeoff is explicitly known.

C. The Gaussian setting

Consider the standard bivariate Gaussian distribution

$$\pi_\rho = \mathcal{N}(\mathbf{0}, \Sigma_\rho),$$

where $0 \leq \rho < 1$ and

$$\Sigma_\rho = \begin{pmatrix} 1 & \rho \\ \rho & 1 \end{pmatrix}.$$

For this distribution, Wyner's common information is

$$C_{\text{Wyner}}(\pi_\rho) = \frac{1}{2} \ln \frac{1+\rho}{1-\rho}.$$

Indeed, it is achieved by the Gaussian decomposition

$$W \sim \mathcal{N}(0, \rho), \quad X = W + N_X, \quad Y = W + N_Y,$$

where

$$N_X, N_Y \sim \mathcal{N}(0, 1 - \rho)$$

are independent of each other and of W .

As already mentioned, the exact common information is always at least Wyner's common information. Yu and Tan [3] proved the upper bound

$$C_{\text{Exact}}(\pi_\rho) \leq \frac{1}{2} \ln \frac{1+\rho}{1-\rho} + \frac{\rho}{1+\rho}. \quad (3)$$

They also proved that the exact common information equals the ∞ -Rényi common information for this Gaussian source. They conjectured this upper bound is tight.

Conjecture 1 ([3]). *For every $0 \leq \rho < 1$,*

$$C_{\text{Exact}}(\pi_\rho) = \frac{1}{2} \ln \frac{1+\rho}{1-\rho} + \frac{\rho}{1+\rho}.$$

The importance of this question is that the second term $\frac{\rho}{1+\rho}$ is strictly positive whenever $\rho > 0$. Hence the conjecture would show that exact synthesis requires strictly more common randomness than Wyner's approximate synthesis in the Gaussian setting.

As for exact channel synthesis of Gaussian distributions, Yu and Tan [13] proved the inner bound

$$\mathcal{R}_{\text{Exact}}(\pi_\rho) \supseteq \mathcal{R}(\pi_\rho), \quad (4)$$

where

$$\mathcal{R}(\pi_\rho) := \left\{ \begin{array}{ll} (R, R_0) & : \alpha \in [\rho^2, 1], \alpha\beta = \rho^2, \\ R & \geq \frac{1}{2} \log \left[\frac{1}{1-\alpha} \right], \\ R + R_0 & \geq \frac{1}{2} \log \left[\frac{1-\rho^2}{(1-\alpha)(1-\beta)} \right] + \frac{\rho\sqrt{(1-\alpha)(1-\beta)}}{1-\rho^2} \end{array} \right\}.$$

They conjectured this inner bound is tight.

Conjecture 2 ([13]). *For every $0 \leq \rho < 1$,*

$$\mathcal{R}_{\text{Exact}}(\pi_\rho) = \mathcal{R}(\pi_\rho).$$

D. Main results

Our first main contribution is the following lower bound on $C_{\text{Exact}}(\pi_\rho)$. Recall $\pi_\rho = \mathcal{N}(\mathbf{0}, \Sigma_\rho)$.

Theorem 1. *For every $0 \leq \rho < 1$,*

$$C_{\text{Exact}}(\pi_\rho) \geq \frac{1}{2} \log \frac{1+\rho}{1-\rho} + \frac{\rho}{1+\rho}.$$

Combining this lower bound with Yu–Tan's upper bound in (3) confirm Conjecture 1 positively.

Theorem 2. *For every $0 \leq \rho < 1$,*

$$C_{\text{Exact}}(\pi_\rho) = \frac{1}{2} \ln \frac{1+\rho}{1-\rho} + \frac{\rho}{1+\rho}.$$

Remark 1. If $\rho < 0$, replace Y by $-Y$. Since exact common information is invariant under deterministic bijections applied separately to either terminal, $C_{\text{Exact}}(\pi_\rho) = C_{\text{Exact}}(\pi_{|\rho|})$. Hence the general scalar Gaussian formula is

$$C_{\text{Exact}}(\pi_\rho) = \frac{1}{2} \log \frac{1+|\rho|}{1-|\rho|} + \frac{|\rho|}{1+|\rho|}, \quad |\rho| < 1.$$

This theorem states that

$$\text{exact CI} = \text{Wyner CI} + \text{exactness penalty},$$

with

$$\text{exactness penalty} = C_{\text{Exact}}(\pi_\rho) - C_{\text{Wyner}}(\pi_\rho) = \frac{\rho}{1+\rho}.$$

Thus the exact common information is strictly larger than Wyner's common information for every $\rho > 0$. The gap satisfies

$$0 < \frac{\rho}{1+\rho} < \frac{1}{2},$$

and

$$\lim_{\rho \uparrow 1} \frac{\rho}{1+\rho} = \frac{1}{2}.$$

In bits per source symbol, the gap is

$$\frac{\rho}{1+\rho} \log_2 e \leq \frac{1}{2} \log_2 e \approx 0.7213.$$

The proof idea of Theorem 1 can be applied to exact channel synthesis of Gaussian distributions, leading to our second main contribution.

Theorem 3. *For every $0 \leq \rho < 1$,*

$$\mathcal{R}_{\text{Exact}}(\pi_\rho) \subseteq \mathcal{R}(\pi_\rho).$$

Combining this outer bound with Yu–Tan's inner bound in (4) confirm Conjecture 2 positively.

Theorem 4. *For every $0 \leq \rho < 1$,*

$$\mathcal{R}_{\text{Exact}}(\pi_\rho) = \mathcal{R}(\pi_\rho). \tag{5}$$

II. PROOF OF THEOREM 1

In this section, we prove Theorem 1 by using the following strategy:

$$\begin{aligned} \text{exact synthesis} &\implies \text{multiletter bound} \implies \text{optimal transport} \\ &\implies \text{Gaussian } T_2 \implies \text{covariance determinant extremality.} \end{aligned}$$

The case $\rho = 0$ is trivial, and thus, we only consider $\rho \in (0, 1)$.

A. A Multi-letter Bound

Define

$$\pi_\rho^{\otimes n} = \mathcal{N}(\mathbf{0}, \Sigma_\rho^{\otimes n}),$$

where

$$\Sigma_\rho^{\otimes n} = \begin{pmatrix} I_n & \rho I_n \\ \rho I_n & I_n \end{pmatrix}.$$

Equivalently, $X^n \sim \mathcal{N}(0, I_n)$, $Y^n \sim \mathcal{N}(0, I_n)$, and $\mathbb{E}[X^n(Y^n)^\top] = \rho I_n$. We also use $\pi_\rho^{\otimes n}$ to denote the corresponding density.

For two probability measures μ, ν on $\mathbb{R}^n$, let $\mathcal{C}(\mu, \nu)$ denote the set of all couplings of μ and ν . Define the maximal Gaussian cross-entropy

$$\mathcal{H}_\rho^{(n)}(\mu, \nu) := \sup_{Q \in \mathcal{C}(\mu, \nu)} \int Q(dx, dy) \log \frac{1}{\pi_\rho^{\otimes n}(x^n, y^n)}. \quad (6)$$

We now introduce the n -letter functional:

$$\Gamma_n(\rho) := \inf \left\{ -h(X^n|W) - h(Y^n|W) + \mathbb{E}_W \mathcal{H}_\rho^{(n)}(P_{X^n|W}, P_{Y^n|W}) \right\}, \quad (7)$$

where the infimum is over all discrete W and conditional distributions satisfying $P_{X^n Y^n} = \pi_\rho^{\otimes n}$, $X^n - W - Y^n$.

Because of conditional independence,

$$h(X^n, Y^n|W) = h(X^n|W) + h(Y^n|W).$$

We first prove the fundamental converse.

Proposition 1. *For every exact n -letter synthesis code $(W, P_{X^n|W}, P_{Y^n|W})$ for $\pi_\rho^{\otimes n}$, we have $H(W) \geq \Gamma_n(\rho)$. As a consequence,*

$$C_{\text{Exact}}(\pi_\rho) \geq \limsup_{n \rightarrow \infty} \frac{1}{n} \Gamma_n(\rho).$$

Proof: Fix w with $P_W(w) > 0$. Since

$$\pi_\rho^{\otimes n}(x^n, y^n) = \sum_{w'} P_W(w') p_{X^n|w'}(x^n) p_{Y^n|w'}(y^n)$$

with $p_{X^n|w'}, p_{Y^n|w'}$ denote the densities of $P_{X^n|W=w'}, P_{Y^n|W=w'}$, we have pointwise

$$\pi_\rho^{\otimes n}(x^n, y^n) \geq P_W(w) p_{X^n|w}(x^n) p_{Y^n|w}(y^n). \quad (8)$$

Taking logarithms,

$$\log \frac{1}{\pi_\rho^{\otimes n}(x^n, y^n)} \leq -\log P_W(w) - \log p_{X^n|w}(x^n) - \log p_{Y^n|w}(y^n).$$

Let $Q_w \in \mathcal{C}(P_{X^n|w}, P_{Y^n|w})$ be arbitrary. Integrating with respect to Q_w gives

$$\int Q_w(dx, dy) \log \frac{1}{\pi_\rho^{\otimes n}(x^n, y^n)} \leq -\log P_W(w) + h(X^n|W=w) + h(Y^n|W=w).$$

Since the inequality holds for every coupling Q_w ,

$$\mathcal{H}_\rho^{(n)}(P_{X^n|w}, P_{Y^n|w}) \leq -\log P_W(w) + h(X^n|w) + h(Y^n|w).$$

Therefore,

$$-h(X^n|w) - h(Y^n|w) + \mathcal{H}_\rho^{(n)}(P_{X^n|w}, P_{Y^n|w}) \leq -\log P_W(w).$$

Averaging over W ,

$$-h(X^n|W) - h(Y^n|W) + \mathbb{E} \mathcal{H}_\rho^{(n)}(P_{X^n|W}, P_{Y^n|W}) \leq H(W).$$

Taking the infimum over all exact decompositions proves $H(W) \geq \Gamma_n(\rho)$. ■

B. Evaluation of the Maximal Gaussian Cross-Entropy

We now evaluate the functional in (7). Fix w , and write

$$a_w = \mathbb{E}[X^n|W=w], \quad b_w = \mathbb{E}[Y^n|W=w],$$

and

$$A_w = \text{Cov}(X^n|W=w), \quad B_w = \text{Cov}(Y^n|W=w).$$

Let

$$\mu_w^0 = \mathcal{L}(X^n - a_w|W=w), \quad \nu_w^0 = \mathcal{L}(Y^n - b_w|W=w).$$

The Gaussian density is

$$\pi_\rho^{\otimes n}(x^n, y^n) = \frac{1}{(2\pi)^n(1-\rho^2)^{n/2}} \exp \left\{ -\frac{|x^n|^2 + |y^n|^2 - 2\rho\langle x^n, y^n \rangle}{2(1-\rho^2)} \right\},$$

where $|\cdot|$ is the Euclidian norm. Hence,

$$-\log \pi_\rho^{\otimes n}(x^n, y^n) = n \log(2\pi\sqrt{1-\rho^2}) + \frac{|x^n|^2 + |y^n|^2 - 2\rho\langle x^n, y^n \rangle}{2(1-\rho^2)}. \quad (9)$$

Since the marginals of the coupling are fixed, maximizing the cross-entropy is equivalent, for $\rho \geq 0$, to minimizing $\mathbb{E}\langle X^n, Y^n \rangle$.

Lemma 1 (Optimal transport representation). *For every w ,*

$$\inf_{Q \in \mathcal{C}(P_{X^n|w}, P_{Y^n|w})} \mathbb{E}_Q \langle X^n, Y^n \rangle = a_w \cdot b_w + \frac{1}{2} [W_2^2(\mu_w^0, -\nu_w^0) - \text{tr } A_w - \text{tr } B_w]. \quad (10)$$

Proof: Under any coupling,

$$\begin{aligned}\mathbb{E}\langle X^n, Y^n \rangle &= a_w \cdot b_w + \mathbb{E}\langle X^n - a_w, Y^n - b_w \rangle \\ &= a_w \cdot b_w + \frac{1}{2} \mathbb{E}|X^n - a_w + Y^n - b_w|^2 - \frac{1}{2} \text{tr } A_w - \frac{1}{2} \text{tr } B_w.\end{aligned}$$

Minimizing $\mathbb{E}\langle X^n, Y^n \rangle$ is therefore equivalent to minimizing $\mathbb{E}|X^n - a_w + Y^n - b_w|^2$. But $X^n - a_w$ has law μ_w^0 , while $-(Y^n - b_w)$ has law $-\nu_w^0$. Hence the minimum of $\mathbb{E}|X^n - a_w + Y^n - b_w|^2$ is $W_2^2(\mu_w^0, -\nu_w^0)$. $\blacksquare$

Combining the lemma with (9) gives the following exact expression.

Proposition 2. *For every w ,*

$$\begin{aligned}\mathcal{H}_\rho^{(n)}(P_{X^n|w}, P_{Y^n|w}) &= n \log(2\pi \sqrt{1 - \rho^2}) \\ &+ \frac{|a_w|^2 + |b_w|^2 + (1 + \rho)(\text{tr } A_w + \text{tr } B_w) - 2\rho a_w \cdot b_w - \rho W_2^2(\mu_w^0, -\nu_w^0)}{2(1 - \rho^2)}.\end{aligned}\quad (11)$$

C. Fathi's Gaussian Transport Inequality

We next need to control the Wasserstein term in (11). We use Fathi's Gaussian T_2 inequality.

Lemma 2 (Fathi's Gaussian transport inequality [14]). *Let μ, ν be probability measures on $\mathbb{R}^n$ with finite second moments, and let $\gamma = \mathcal{N}(\mathbf{0}, I_n)$. If μ is centered, then*

$$W_2^2(\mu, \nu) \leq 2D(\mu\|\gamma) + 2D(\nu\|\gamma).$$

More generally, applying the scaling transformation $X \mapsto \sqrt{t}X$ to the underlying random variables for all μ, ν, γ (and hence scaling their distributions accordingly), we obtain

$$W_2^2(\mu, \nu) \leq 2t[D(\mu\|\gamma_t) + D(\nu\|\gamma_t)], \quad (12)$$

where $\gamma_t = \mathcal{N}(0, tI_n)$.

This scaling is exactly what we need. We fix temporarily $t \in (0, \frac{1-\rho^2}{\rho})$ (and lastly, will choose $t = 1 - \rho$). Since μ_w^0 and $-\nu_w^0$ are centered, applying (12) to them yields

$$W_2^2(\mu_w^0, -\nu_w^0) \leq S_w - 2tH_w + 2kt \log(2\pi t), \quad (13)$$

where

$$S_w = \text{tr } A_w + \text{tr } B_w$$

and

$$H_w = h(X^n|w) + h(Y^n|w).$$

Substituting (13) into (11) yields

$$\begin{aligned}\mathcal{H}_\rho^{(n)}(P_{X^n|w}, P_{Y^n|w}) &\geq n \log(2\pi \sqrt{1 - \rho^2}) + \frac{|a_w|^2 + |b_w|^2 + S_w - 2\rho a_w \cdot b_w}{2(1 - \rho^2)} \\ &+ \frac{\rho t}{1 - \rho^2} H_w - \frac{\rho k t}{1 - \rho^2} \log(2\pi t).\end{aligned}\quad (14)$$

Consequently, the contribution of w to $\Gamma_n(\rho)$, denoted by g , satisfies

$$g(w) \geq n \log(2\pi\sqrt{1-\rho^2}) + \frac{|a_w|^2 + |b_w|^2 + S_w - 2\rho a_w \cdot b_w}{2(1-\rho^2)} + \left(-1 + \frac{\rho t}{1-\rho^2}\right) H_w - \frac{\rho k t}{1-\rho^2} \log(2\pi t). \quad (15)$$

Since $(X^n, Y^n) \sim \pi_\rho^{\otimes n}$, we have $\mathbb{E}|X^n|^2 = \mathbb{E}|Y^n|^2 = n$, and $\mathbb{E}[X^n \cdot Y^n] = n\rho$. By the law of total covariance,

$$I_n = \mathbb{E}[A_W] + \text{Cov}(a_W),$$

and therefore

$$\mathbb{E} [\text{tr } A_W + |a_W|^2] = n.$$

Similarly,

$$\mathbb{E} [\text{tr } B_W + |b_W|^2] = n.$$

Finally, conditional independence gives $\text{Cov}(X^n, Y^n) = \text{Cov}(a_W, b_W)$, so $\mathbb{E}[a_W \cdot b_W] = n\rho$.

Hence,

$$\mathbb{E} [|a_W|^2 + |b_W|^2 + S_W - 2\rho a_W \cdot b_W] = 2k(1-\rho^2).$$

Averaging (15) therefore gives

$$\Gamma_n(\rho) \geq n \log(2\pi\sqrt{1-\rho^2}) + n + \left(-1 + \frac{\rho t}{1-\rho^2}\right) \mathbb{E} H_W - \frac{\rho k t}{1-\rho^2} \log(2\pi t). \quad (16)$$

At this point, all source-specific information has been reduced to an upper bound on the conditional entropy $\mathbb{E} H_W$.

D. Entropy Bound from the Conditional Covariance

Define

$$U = \text{Cov}(\mathbb{E}[X^n|W]) = \text{Cov}(a_W),$$

and

$$V = \text{Cov}(\mathbb{E}[Y^n|W]) = \text{Cov}(b_W).$$

Since $\text{Cov}(X^n, Y^n) = \rho I_n$ and conditional independence gives $\text{Cov}(X^n, Y^n) = \text{Cov}(a_W, b_W)$, we obtain $\text{Cov}(a_W, b_W) = \rho I_n$. Thus,

$$\begin{pmatrix} U & \rho I_n \\ \rho I_n & V \end{pmatrix} \succeq 0. \quad (17)$$

Moreover,

$$I_n - U = \mathbb{E} \text{Cov}(X^n|W) \succeq 0,$$

and similarly

$$I_n - V \succeq 0.$$

We now prove the determinant inequality needed for the entropy estimate.

Lemma 3. For $0 < \rho < 1$,

$$\det(I_n - U) \det(I_n - V) \leq (1 - \rho)^{2k}.$$

Proof: For $\rho > 0$, the positive semidefiniteness in (17) implies that U is positive definite. Taking the Schur complement of U , $V - \rho^2 U^{-1} \succeq 0$. Hence $V \succeq \rho^2 U^{-1}$. Since $V \preceq I_n$, we have $\rho^2 U^{-1} \preceq I_n$. Thus every eigenvalue u_i of U satisfies

$$\rho^2 \leq u_i \leq 1. \quad (18)$$

From $V \succeq \rho^2 U^{-1}$, we obtain $I_n - V \preceq I_n - \rho^2 U^{-1}$. Therefore,

$$\det(I_n - V) \leq \det(I_n - \rho^2 U^{-1}).$$

Let $u_1, \dots, u_n$ be the eigenvalues of U . Then,

$$\det(I_n - U) \det(I_n - V) \leq \det(I_n - U) \det(I_n - \rho^2 U^{-1}) = \prod_{i=1}^n (1 - u_i) \left(1 - \frac{\rho^2}{u_i}\right). \quad (19)$$

For every $u \in [\rho^2, 1]$,

$$(1 - \rho)^2 - (1 - u) \left(1 - \frac{\rho^2}{u}\right) = \frac{(u - \rho)^2}{u} \geq 0.$$

Consequently,

$$(1 - u_i) \left(1 - \frac{\rho^2}{u_i}\right) \leq (1 - \rho)^2$$

for every i , which gives

$$\det(I_n - U) \det(I_n - V) \leq (1 - \rho)^{2k}.$$

■

Lemma 4 (Conditional entropy bound). For every exact Gaussian code,

$$\mathbb{E}H_W \leq n \log(2\pi e(1 - \rho)).$$

Proof: For each w , Gaussian maximal entropy gives

$$h(X^n|w) \leq \frac{n}{2} \log(2\pi e) + \frac{1}{2} \log \det A_w.$$

Averaging and using concavity of $\log \det$,

$$\begin{aligned} h(X^n|W) &\leq \frac{n}{2} \log(2\pi e) + \frac{1}{2} \log \det \mathbb{E}[A_W] \\ &= \frac{n}{2} \log(2\pi e) + \frac{1}{2} \log \det(I_n - U). \end{aligned}$$

Similarly,

$$h(Y^n|W) \leq \frac{n}{2} \log(2\pi e) + \frac{1}{2} \log \det(I_n - V).$$

Therefore

$$\mathbb{E}H_W \leq n \log(2\pi e) + \frac{1}{2} \log [\det(I_n - U) \det(I_n - V)]. \quad (20)$$

Applying the determinant lemma,

$$\mathbb{E}H_W \leq n \log(2\pi e) + n \log(1 - \rho) = n \log(2\pi e(1 - \rho)).$$

■

E. Optimization over the Transport Parameter

Substituting the conditional entropy bound into (16), we obtain

$$\begin{aligned} \Gamma_n(\rho) &\geq n \log(2\pi e \sqrt{1 - \rho^2}) \\ &\quad + \left(-1 + \frac{\rho t}{1 - \rho^2}\right) n \log(2\pi e(1 - \rho)) - \frac{\rho k t}{1 - \rho^2} \log(2\pi t). \end{aligned} \quad (21)$$

For $0 < \rho < 1$, the derivative of the right-hand side with respect to t is

$$-\frac{\rho n}{1 - \rho^2} \log \frac{t}{1 - \rho}.$$

Hence the unique maximizer is $t = 1 - \rho$. Substituting $t = 1 - \rho$ gives

$$\Gamma_n(\rho) \geq n \left[\frac{1}{2} \log \frac{1 + \rho}{1 - \rho} + \frac{\rho}{1 + \rho} \right]. \quad (22)$$

By the multi-letter converse and (22),

$$C_{\text{Exact}}(\pi_\rho) \geq \limsup_{n \rightarrow \infty} \frac{\Gamma_n(\rho)}{n} \geq \frac{1}{2} \log \frac{1 + \rho}{1 - \rho} + \frac{\rho}{1 + \rho}.$$

III. PROOF OF THEOREM 3

The case $\rho = 0$ is trivial, and thus, we only consider $\rho \in (0, 1)$.

A. A Multi-letter Bound

We now introduce the n -letter rate region:

$$\mathcal{R}_n(\pi_\rho) := \left\{ \begin{array}{ll} (R_0, R) & : \exists P_W P_{X^n|W} P_{Y^n|W} \text{ s.t.} \\ P_{X^n Y^n} & = \pi_\rho^{\otimes n}, \\ R & \geq \frac{1}{n} I(W; X^n), \\ R + R_0 & \geq \frac{1}{n} \Gamma_n(P_W, P_{X^n|W}, P_{Y^n|W}) \end{array} \right\}, \quad (23)$$

where

$$\Gamma_n(P_W, P_{X^n|W}, P_{Y^n|W}) := -h(X^n|W) - h(Y^n|W) + \mathbb{E}_W \mathcal{H}_\rho^{(n)}(P_{X^n|W}, P_{Y^n|W}), \quad (24)$$

which is the objective function in (7). We first prove the fundamental converse.

Proposition 3. *For every exact n -letter channel synthesis code $(R_0, P_{M|X^n K}, P_{Y^n|MK})$ for $\pi_\rho^{\otimes n}$ and letting $W = (M, K)$, we have $X^n \rightarrow W \rightarrow Y^n$, $R \geq \frac{1}{n} I(W; X^n)$, and $R_0 + R \geq \frac{1}{n} \Gamma_n(P_W, P_{X^n|W}, P_{Y^n|W})$. As a consequence,*

$$\mathcal{R}_{\text{Exact}}(\pi_\rho) \subseteq \text{cl} \bigcup_{n \geq 1} \mathcal{R}_n(\pi_\rho).$$

Proof: By examining the proof of Proposition 1,

$$kR + kR_0 \geq H(W) \geq -h(X^n|W) - h(Y^n|W) + \mathbb{E} \mathcal{H}_\rho^{(n)}(P_{X^n|W}, P_{Y^n|W}).$$

Moreover,

$$kR \geq H(M|K) \geq I(X^n; M|K) = I(X^n; M, K) = I(X^n; W).$$

■

B. Evaluation of the Multi-letter Bound

Recall that

$$U = \text{Cov}(\mathbb{E}[X^n|W]) = \text{Cov}(a_W),$$

and

$$V = \text{Cov}(\mathbb{E}[Y^n|W]) = \text{Cov}(b_W).$$

Let $u_i, 1 \leq i \leq n$ be eigenvalues of U . From (18), we know $\rho^2 \leq u_i \leq 1$.

From (20) and (19), we get

$$\begin{aligned} \mathbb{E} H_W &\leq n \log(2\pi e) + \frac{1}{2} \log [\det(I_n - U) \det(I_n - V)] \\ &= n \log(2\pi e) + \frac{1}{2} \sum_{i=1}^n \log \left[(1 - u_i) \left(1 - \frac{\rho^2}{u_i} \right) \right]. \end{aligned} \quad (25)$$

Moreover, we have

$$\begin{aligned}
I(X^n; W) &= h(X^n) - h(X^n|W) \\
&\geq \frac{n}{2} \log(2\pi e) - \mathbb{E}\left[\frac{1}{2} \log\{(2\pi e)^n \det(\text{Cov}(X^n | W))\}\right] \\
&= -\frac{1}{2} \mathbb{E}[\log \det(A_W)] \\
&\geq -\frac{1}{2} \log \det(\mathbb{E}[A_W]) \\
&= -\frac{1}{2} \log \det(I_n - U) \\
&= -\frac{1}{2} \sum_{i=1}^n \log(1 - u_i),
\end{aligned}$$

where the first inequality follows since Gaussian distributions maximize entropy, the second inequality follows by the concavity of $\log \det$, and the third equality follows by the law of total covariance.

We next show that setting $u_i, 1 \leq i \leq n$ to the same value will lead to a further bound.

Lemma 5. *Assume that $\rho^2 \leq u_i \leq 1$ for all i . If*

$$\frac{1}{n} \sum_{i=1}^n \log(1 - u_i) = \log(1 - \alpha),$$

then

$$\frac{1}{n} \sum_{i=1}^n \log\left(1 - \frac{\rho^2}{u_i}\right) \leq \log\left(1 - \frac{\rho^2}{\alpha}\right).$$

Proof: Define $x_i = \log(1 - u_i)$, so that the assumption is equivalent to

$$\frac{1}{n} \sum_{i=1}^n x_i = \log(1 - \alpha).$$

Consider the function

$$f(x) = \log\left(1 - \frac{\rho^2}{1 - e^x}\right),$$

defined for $x \leq \log(1 - \rho^2)$. Since $u = 1 - e^x$, we have

$$f(\log(1 - u)) = \log\left(1 - \frac{\rho^2}{u}\right).$$

We first show that h is concave. A direct calculation gives

$$f''(x) = e^x \left(\frac{1}{(1 - e^x)^2} - \frac{1 - \rho^2}{(1 - \rho^2 - e^x)^2} \right).$$

Because $u = 1 - e^x \geq \rho^2$, we have $e^x \leq 1 - \rho^2$. Moreover,

$$(1 - \rho^2 - e^x)^2 \leq (1 - \rho^2)(1 - e^x)^2,$$

which implies

$$\frac{1}{(1 - e^x)^2} \leq \frac{1 - \rho^2}{(1 - \rho^2 - e^x)^2}.$$

Hence $f''(x) \leq 0$, and therefore f is concave.

By Jensen's inequality for concave functions,

$$\frac{1}{n} \sum_{i=1}^n f(x_i) \leq f\left(\frac{1}{n} \sum_{i=1}^n x_i\right).$$

Using the definition of x_i and the constraint, we obtain

$$\frac{1}{n} \sum_{i=1}^n \log\left(1 - \frac{\rho^2}{u_i}\right) \leq \log\left(1 - \frac{\rho^2}{\alpha}\right).$$

■

Applying Lemma 5 yields that for some α such that $\rho^2 \leq \alpha \leq 1$, it holds that

$$I(X^n; W) \geq -\frac{1}{2} \log(1 - \alpha),$$

and

$$\frac{\mathbb{E}H_W}{n} \leq \log(2\pi e) + \frac{1}{2} \log\left[(1 - \alpha) \left(1 - \frac{\rho^2}{\alpha}\right)\right]. \quad (26)$$

Analogously to (16), we have

$$R + R_0 \geq \log(2\pi e \sqrt{1 - \rho^2}) + \left(-1 + \frac{\rho t}{1 - \rho^2}\right) \frac{\mathbb{E}H_W}{n} - \frac{\rho t}{1 - \rho^2} \log(2\pi t).$$

Substituting (26) into this inequality yields

$$\begin{aligned} R + R_0 &\geq \log(2\pi e \sqrt{1 - \rho^2}) - \frac{\rho t}{1 - \rho^2} \log(2\pi t) \\ &\quad + \left(-1 + \frac{\rho t}{1 - \rho^2}\right) \left(\log(2\pi e) + \frac{1}{2} \log\left[(1 - \alpha) \left(1 - \frac{\rho^2}{\alpha}\right)\right]\right). \end{aligned} \quad (27)$$

The optimal parameter t is

$$t = \sqrt{(1 - \alpha) \left(1 - \frac{\rho^2}{\alpha}\right)}.$$

Substituting it into (27) yields the desired bound

$$R + R_0 \geq \frac{1}{2} \log \frac{1 - \rho^2}{(1 - \alpha) \left(1 - \frac{\rho^2}{\alpha}\right)} + \frac{\rho \sqrt{(1 - \alpha) \left(1 - \frac{\rho^2}{\alpha}\right)}}{1 - \rho^2}.$$

Acknowledgements

Generative-AI use disclosure: During the preparation of this manuscript, the author used ChatGPT (GPT-5.5-mini model, Think mode) as an auxiliary tool for exploring proof ideas, checking calculations, and improving exposition. All mathematical content, including statements, proofs, and references, was independently verified by the author, who assumes full responsibility for the final manuscript.

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